CREDIT RISK / DECISION INTELLIGENCE

Credit Risk Intelligence Platform

Tabular default prediction + safe NL-to-SQL portfolio querying

307,511

loan applications

8.07%

actual default rate

LIGHTGBM SHAP RULES GROQ SQLGLOT

PROJECT PRESENTATION

02

What the platform actually does

One risk engine, two ways to use it: score an applicant or interrogate the portfolio.

01

Score an applicant

Applicant data — **LightGBM** — **probability** — **risk band** — **decision**

Low / Medium / High

02

Ask the portfolio

“Which age group has the highest default rate?”

natural language — **Groq → SQL** — **sqlglot gate** — **answer**

The platform is not just a classifier. It connects prediction, explanation and governed data access.

From raw applications to model-ready data

The preprocessing layer is where the project makes its most important data-quality decisions.

307,511

applications in the master dataset

SOURCE MATERIAL

01 application_train

02 bureau

03 previous_application

PREPROCESSING

01	**Retiree placeholder**	365243 → null + is_retiree_placeholder
02	**Missing values**	median imputation
03	**Numeric variables**	standardization
04	**Categoricals**	one-hot encoding
05	**Split**	stratified 80 / 20

178 model features after preprocessing

The 365243 value is treated as a placeholder, not as a real employment duration.

04

A probability becomes an operating decision

LightGBM ranks applicants; the banding layer turns that probability into an action.

MODEL

LightGBM

scale_pos_weight = 11.39

HOLDOUT PERFORMANCE

0.7670	0.2555	0.7602 ± 0.0012
ROC-AUC	PR-AUC	5-fold CV ROC-AUC

LOW	MEDIUM	HIGH
$p < 0.3$	$0.3 \le p < 0.6$	$p \ge 0.6$
40.85% applicant share	39.53%	19.62%
2.28% actual default	7.17%	21.95%
AUTO-APPROVE	**MANUAL REVIEW**	**AUTO-DECLINE**

Scores within 0.05 of a band edge are flagged for borderline human review.

05

The model is not the explanation

Two layers make a probability inspectable: local SHAP for an applicant, and surrogate rules for repeatable business logic.

SHAP

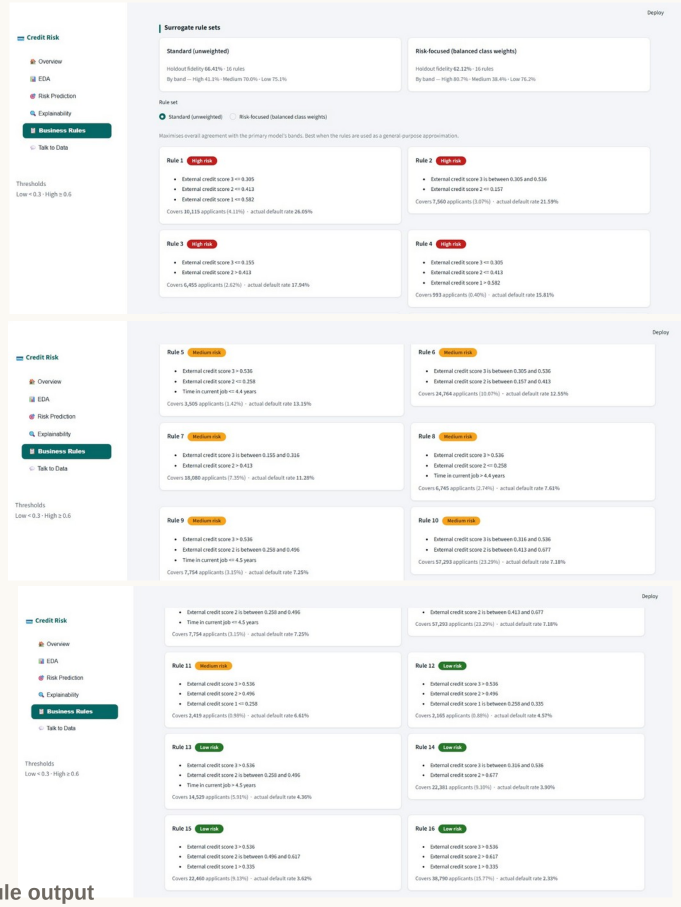
Why this applicant?

Local feature attributions show which variables pushed an individual score higher or lower.

BUSINESS RULES

What pattern does the model learn?

A depth-4 surrogate tree with 16 leaves gives a compact, readable approximation of model behaviour.



actual Streamlit business-rule output

Standard holdout fidelity: 66.41% · risk-focused balanced fidelity: 62.12%

What the data revealed

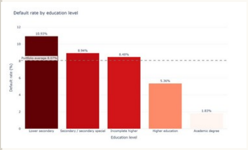
EDA was not decorative. It changed preprocessing and surfaced a fairness issue worth addressing before production.

365,243

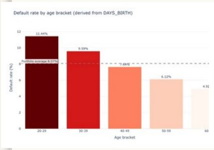
DAYS_EMPLOYED

retiree placeholder

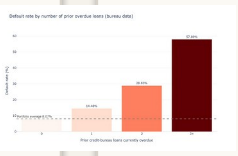
nulled before imputation + binary signal retained



education



age



prior overdue

HIGH-RISK BAND

28.18%

men

15.59%

women

1.8×

difference in High-band placement

WHY IT MATTERS

The current model uses sex and marital status as inputs. The disparity is therefore a production fairness concern, not a claim that the model is “biased” from one metric alone.

NEXT STEP

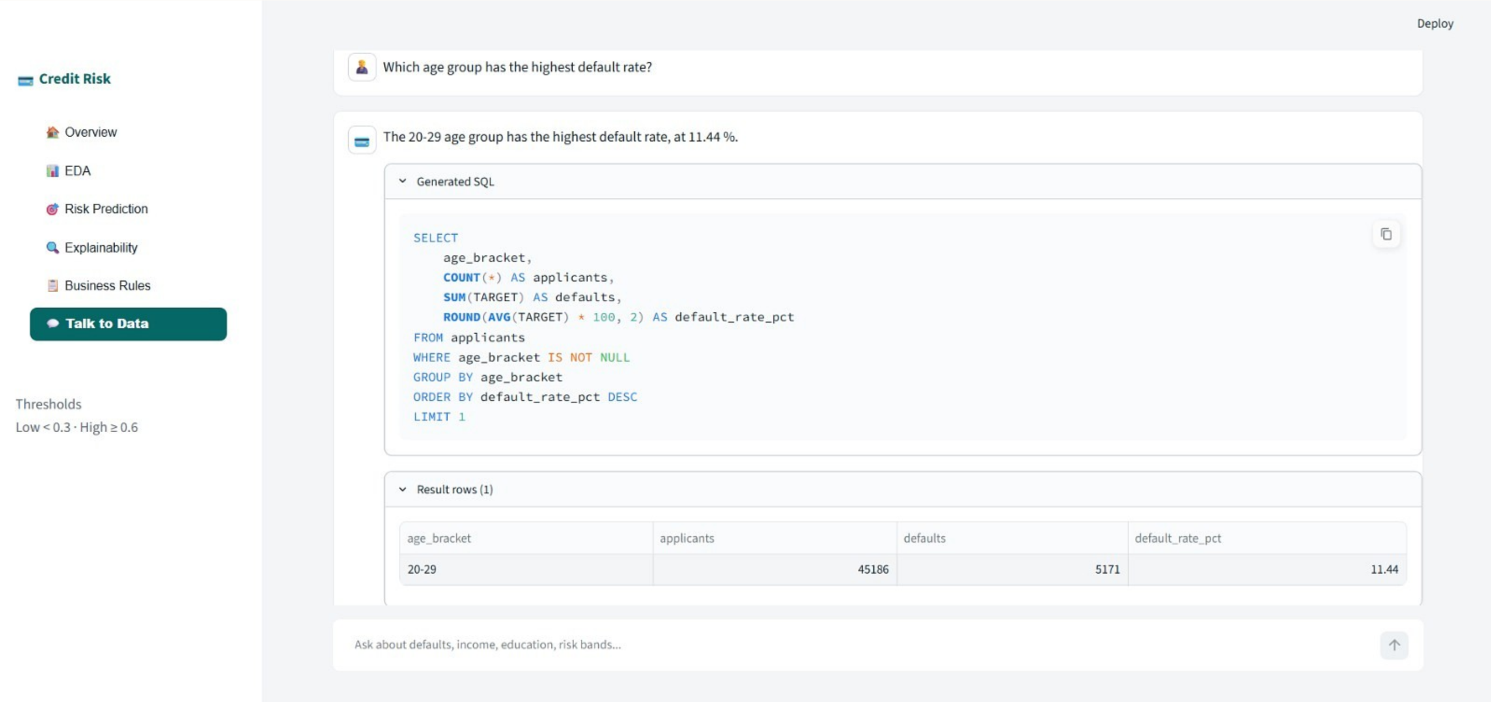
Retrain without protected attributes → compare performance → re-check subgroup outcomes.

Fairness analysis identifies a question for model governance; it does not by itself prove causation.

07

Talk to Data

Natural-language portfolio analysis, with a safety gate between the LLM and the database.



EXAMPLE QUESTIONS

01 Which 5 education types have the highest default rate, considering only groups with at least 1,000 applicants?

02 What is the average income of applicants who have more than 2 prior overdue loans?

03 Compare the default rate of male and female applicants.

The LLM proposes SQL. The application decides whether that SQL is allowed.

THE GOVERNED PATH

01 Question

02 Groq

03 sqlglot

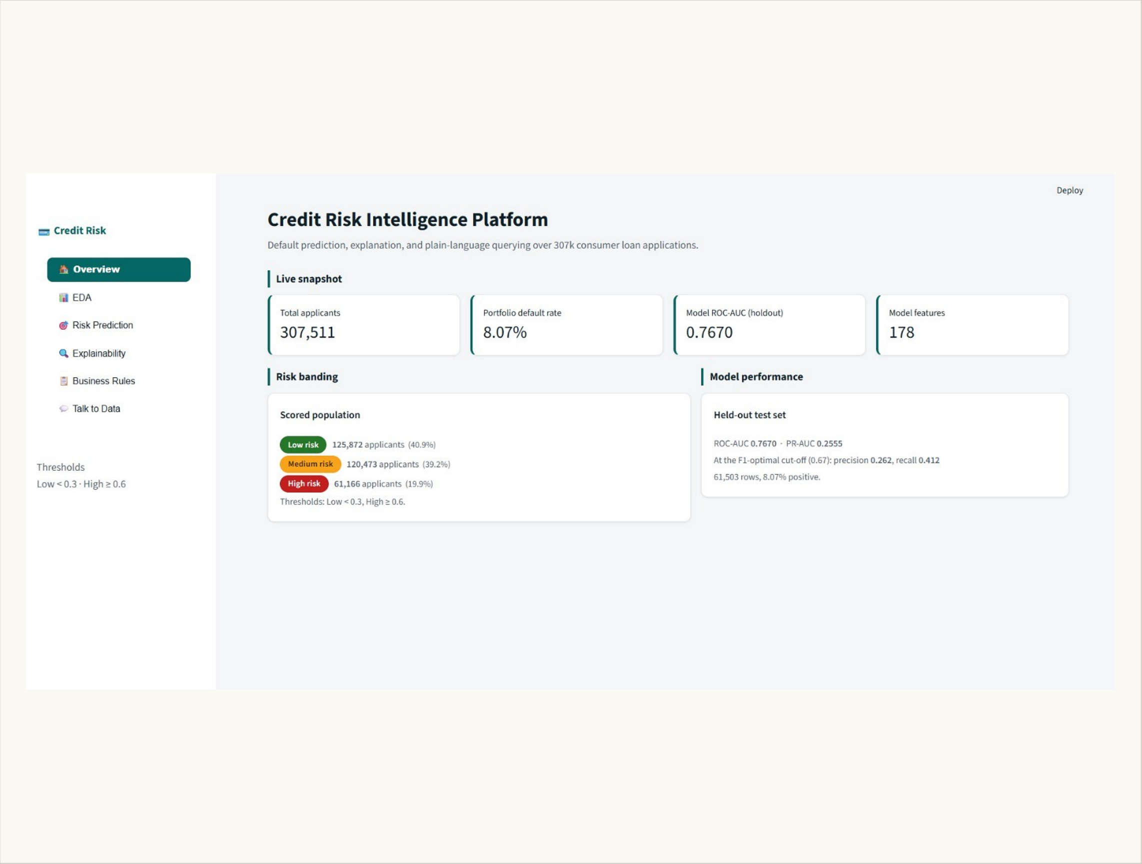
04 SQLite

05 Answer

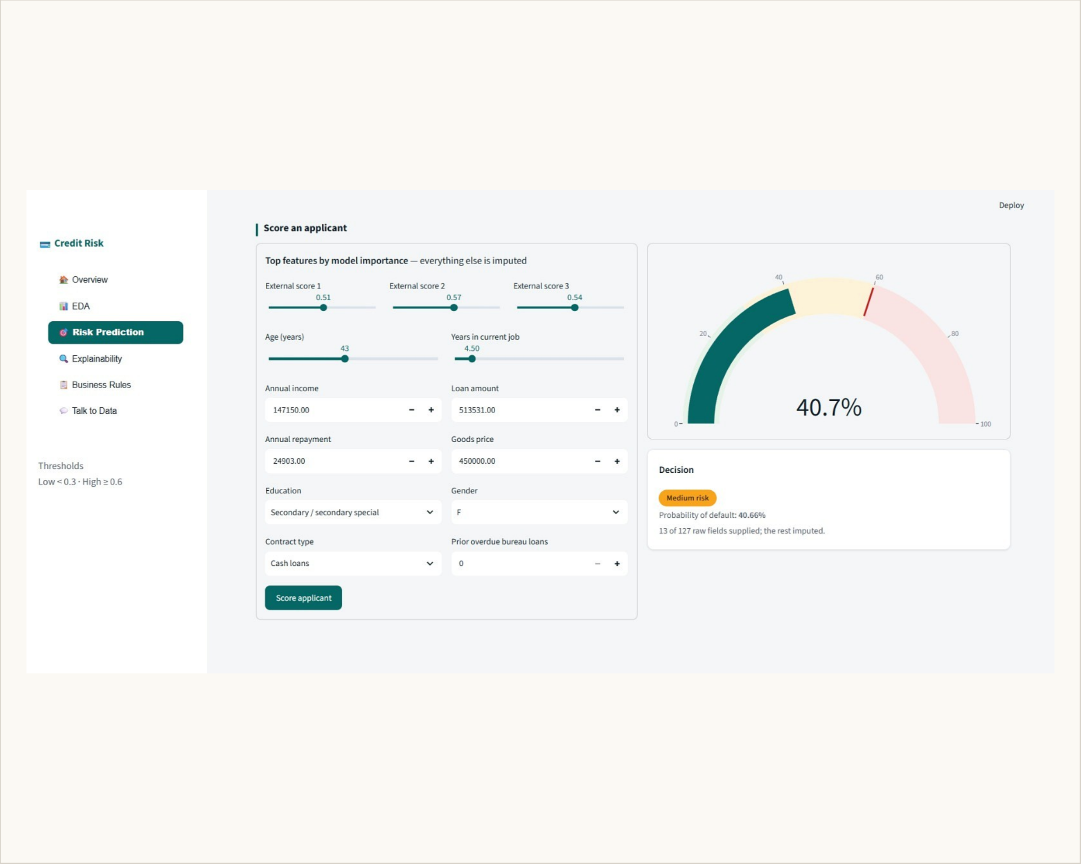
Only one read-only SELECT / UNION path is allowed. Results are capped at 1,000 rows; the second LLM call summarizes returned rows only.

One local interface, three decision views

The Streamlit application brings scoring, explanation, rules and portfolio querying together.



OVERVIEW



RISK PREDICTION

PREDICT → **EXPLAIN** → **QUERY SAFELY**

A credit-risk workflow designed to be inspected, challenged and used.